

Project Requirement Document (PRD)

Project Name: Maintenance Scheduling Tool

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Date: 26-03-2026

1. Introduction

The Maintenance Scheduling Tool is a web-based application designed to manage and track preventive maintenance of equipment used in testing environments such as Test Beds, Chassis Dyno, and Common Facilities.

In the current workflow, maintenance tracking is handled using spreadsheets, which leads to manual effort, limited visibility, and increased chances of missing scheduled maintenance. This system provides a structured interface to define equipment, configure maintenance schedules, and monitor due tasks through a centralized dashboard.

The application ensures that all maintenance activities are clearly visible, organized, and executed on time, improving the reliability and efficiency of testing operations.

2. Objectives

- Provide a centralized platform to manage maintenance scheduling
- Allow users to define equipment and maintenance requirements
- Capture details such as system name, description, commissioning date, and frequency
- Automatically calculate upcoming maintenance schedules
- Display maintenance tasks due in a selected month
- Reduce dependency on manual spreadsheet tracking
- Improve visibility and execution of preventive maintenance

3. Scope

Included :

- Use of predefined environments:
 1. Test Bed
 2. Chassis Dyno
 3. Common Facilities
- Equipment creation under each environment

Input fields:

1. System name
 2. Description
 3. Date of commissioning
 4. Maintenance frequency
- Automatic calculation of next maintenance date
 - Monthly Synthesis dashboard displaying due maintenance tasks
 - Ability to update maintenance completion (updates last maintenance date only)
 - Status identification (Pending / Overdue based on date)

4. Application Overview

The application will be a web-based tool consisting of:

- Equipment management interface categorized by fixed environments
- Maintenance configuration interface
- Synthesis dashboard for monthly maintenance view

Users will define equipment and maintenance rules, and the system will automatically generate and display maintenance tasks based on configured frequency and dates.

5. Technology Stack

- Frontend: React
- Backend: Flask
- Database: SQLite

6. Input Data Source

- User input through equipment and maintenance configuration forms
- Data stored and managed in the database

7. User Flow

1. User opens the application
2. User selects one of the predefined environments:

- Test Bed
 - Chassis Dyno
 - Common Facilities
3. User adds equipment with:
 - System name
 - Description
 - Commissioning date
 - Maintenance frequency
 4. System stores the data
 5. System calculates next maintenance date
 6. User navigates to Synthesis page
 7. System displays:
 - System name
 - Description
 - Next maintenance date (filtered by selected month)
 8. User performs maintenance
 9. User updates maintenance completion (updates last maintenance date)
 10. System recalculates next maintenance date

8. Functional Requirements

8.1 Environment Handling

Purpose: Categorize equipment under predefined environments

Behavior:

- System provides fixed environments:
 1. Test Bed
 2. Chassis Dyno
 3. Common Facilities
- Users cannot create, edit, or delete environments
- Equipment must be assigned to one of these environments

8.2 Equipment Management

Purpose: Store and manage equipment details

Behavior:

- User can add equipment
- Fields include:
 1. System name

2. Description
 3. Environment (selected from predefined list)
 4. Commissioning date
 5. Frequency
- User can edit or delete equipment

8.3 Maintenance Scheduling

Purpose: Automatically determine maintenance dates

Behavior:

- System calculates next maintenance date using:
 1. Commissioning date or last maintenance date
 2. Frequency
- No manual scheduling required

8.4 Synthesis Dashboard

Purpose: Display actionable maintenance tasks

Behavior:

- Displays tasks due in selected month
- Shows:
 1. System name
 2. Description
 3. Next maintenance date
- Allows filtering by:
 1. Environment
 2. Month
- Highlights:
 1. Pending tasks
 2. Overdue tasks

8.5 Maintenance Update

Purpose: Mark maintenance as completed

Behavior:

- User updates maintenance completion
- System updates last maintenance date
- System recalculates next maintenance date

9. Database Design

1. Environments Table (Static / Seeded)

Column	Description
id	Unique identifier
name	Environment name

2. Equipment Table

Column	Description
id	Unique identifier for each equipment record
system_name	Name of the equipment or system
description	Description of the maintenance task to be performed
environment_id	Reference to the predefined environment the equipment belongs to
commissioning_date	Date when the equipment was first put into operation
frequency_type	Unit of maintenance frequency (e.g., daily, weekly, monthly)
frequency_value	Interval value defining how often maintenance should occur
last_maintenance_date	Date when the maintenance was last performed
next_maintenance_date	System-calculated date for the next scheduled maintenance
created_at	Timestamp when the record was created
updated_at	Timestamp when the record was last updated

10. Data Update Rules

- Environments → Predefined and seeded in database
- New equipment → Insert record with commissioning date and frequency
- Maintenance update → Update last maintenance date
- Next maintenance date → Recalculate based on frequency
- Synthesis view → Dynamically filtered based on selected month

11. Security & Access Control

- No role-based restrictions in the initial phase
- All users can create and update data
- Access control to be defined in future phase

12. Page-wise Details

12.1 Synthesis Dashboard Page (Landing Page)

Purpose:

- To display all maintenance tasks due for a selected time period and act as the primary entry point of the application.

Behavior:

- This is the default landing page when the user opens the application
- Displays:
 1. System name
 2. Description
 3. Environment (TB / CD / CF)
 4. Next maintenance date
- Filters available:
 1. Environment (TB / CD / CF / All)
 2. Month / Year
- Status indication:
 1. Pending (upcoming tasks)
 2. Overdue (past due date)
- Provides action to:
 1. Update maintenance completion
 2. Navigate to equipment management page (tabs view)

13.2 Main Page with Environment Tabs

Purpose:

- To manage equipment under predefined environments.

Structure:

- The page consists of three fixed tabs:
 1. Test Bed (TB)
 2. Chassis Dyno (CD)
 3. Common Facilities (CF)

13.3 Equipment Management within Tabs

Purpose:

- To create, view, edit, and manage equipment under each environment.

Behavior:

- When a user selects a tab (TB / CD / CF):
 1. System displays list of equipment under that environment
- Within each tab, user can:
 1. Add new equipment
 2. Edit existing equipment
 3. Delete equipment
- Display fields in list view:
 1. System name
 2. Description
 3. Commissioning date
 4. Frequency
 5. Next maintenance date

13.4 Add / Edit Equipment (Modal)

Purpose:

- To input and update equipment and maintenance configuration details.

Behavior:

- Triggered from within each environment tab
- User enters:
 1. System name

2. Description
 3. Commissioning date
 4. Frequency type
 5. Frequency value
- Environment is automatically assigned based on selected tab
 - On submission:
 1. System validates inputs
 2. Stores data
 3. Calculates next maintenance date

13.5 Maintenance Update Action

Purpose:

- To mark maintenance as completed.

Behavior:

- Available from:
 1. Synthesis dashboard
 2. Equipment list within tabs
- On action:
 1. System updates last maintenance date
 2. System recalculates next maintenance date

13.6 Navigation Structure

Pages Included:

- Synthesis Dashboard (default landing page)
- Main Page (Environment Tabs: TB / CD / CF)

Navigation Behavior:

- User lands on Synthesis Dashboard by default
- From Synthesis, user can navigate to:
 1. Environment Tabs page for equipment management
- From Environment Tabs:
 1. User can manage equipment
 2. User can return to Synthesis dashboard

12. Workflow Diagram

